ECE 440
Lecture 34 : Metal-Semiconductor Junctions

Class Outline:

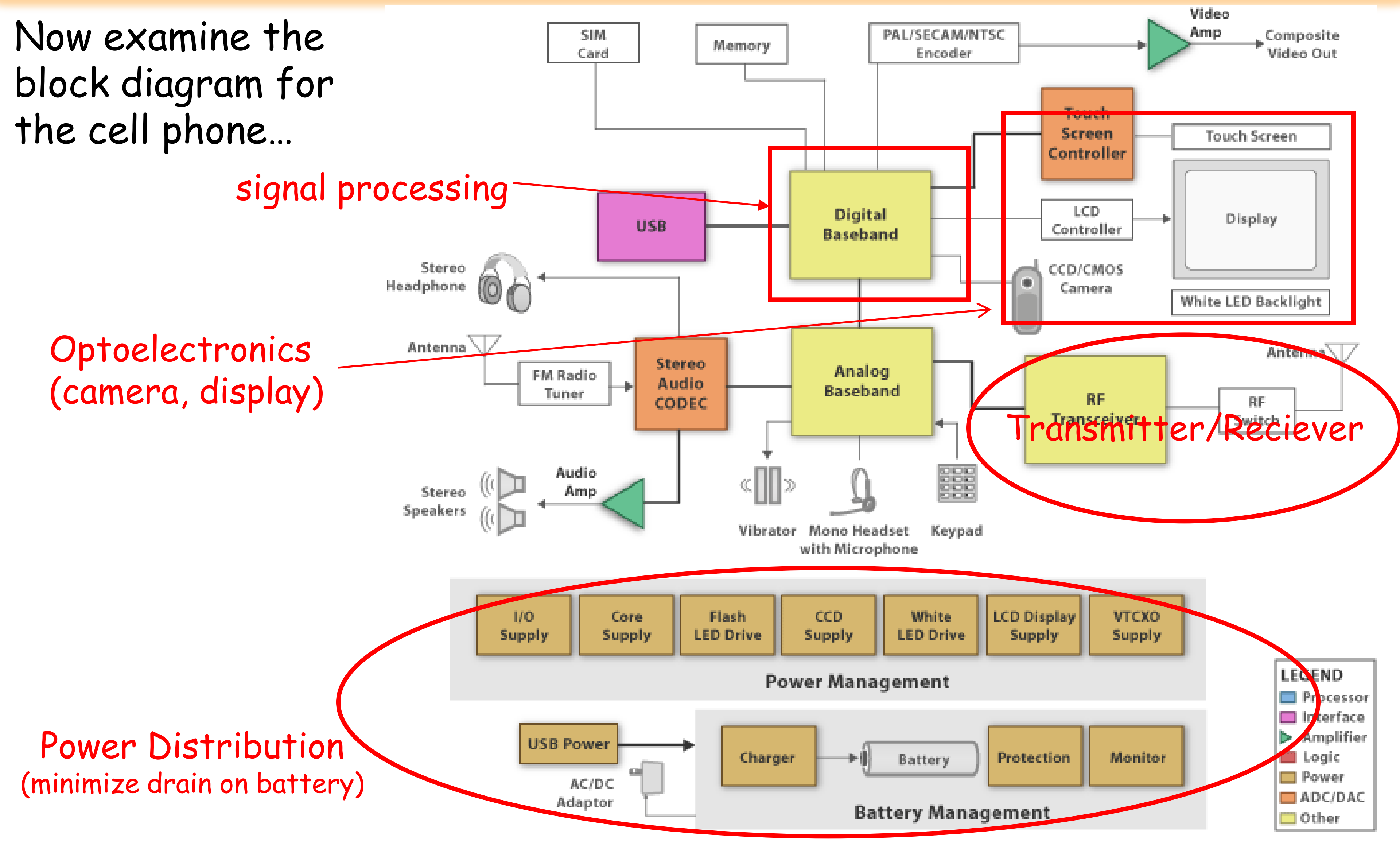
- Ideal Metal-Semiconductor Contacts
- Rectifying Contacts
- Ohmic Contacts

Key Questions

- What happens to the bands when we make contact between metals and semiconductors?
- What is a rectifying contact?
- What is an ohmic contact?
- How does doping change the operation of an ohmic contact?

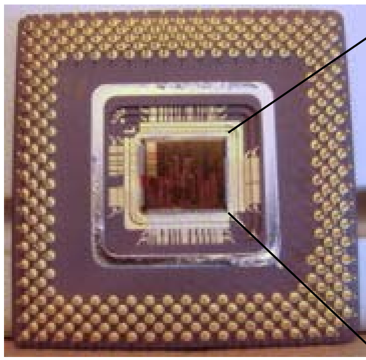
Ideal Metal-Semiconductor Contacts

Now examine the block diagram for the cell phone…

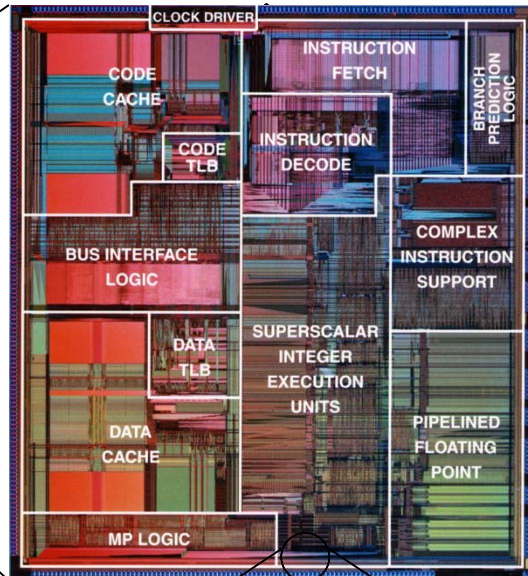


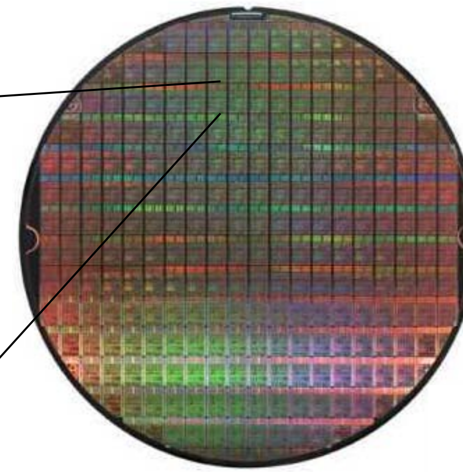
Ideal Metal-Semiconductor Contacts

It contains a microprocessor unit…



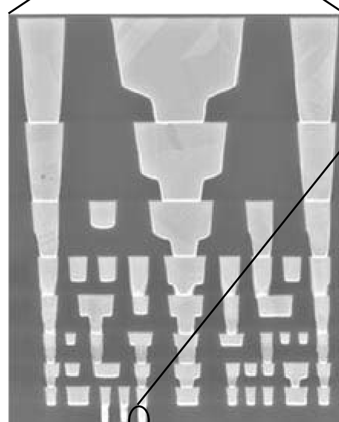
Integrated circuit containing millions of

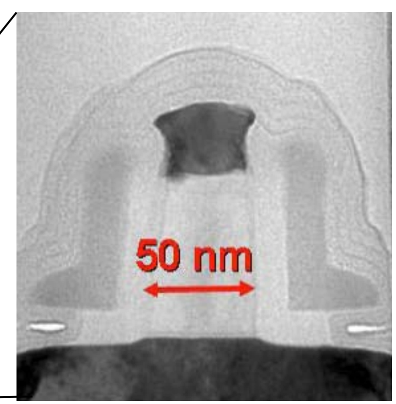




Si wafer containing thousands of ICs

cross section

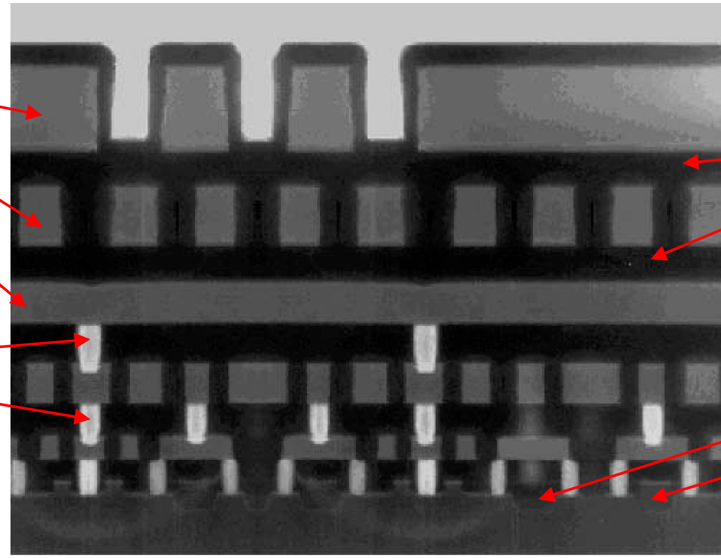


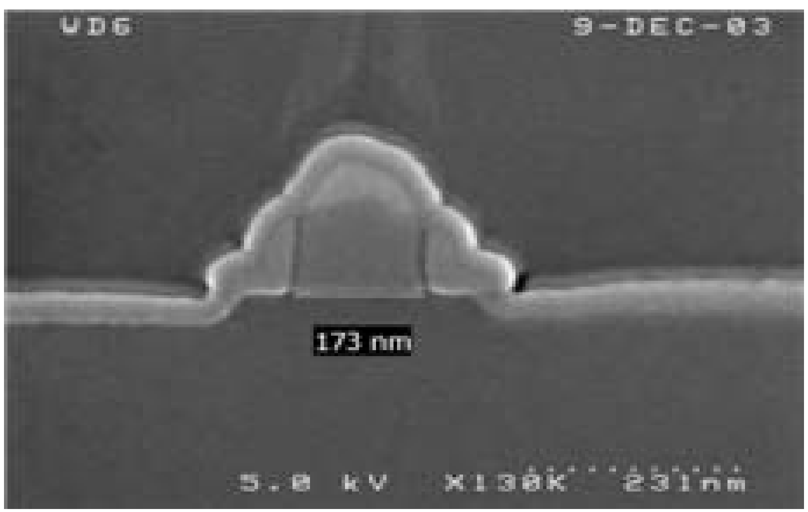


single transistor

Ideal Metal-Semiconductor Contacts

The cross-section has many different layers, but why?



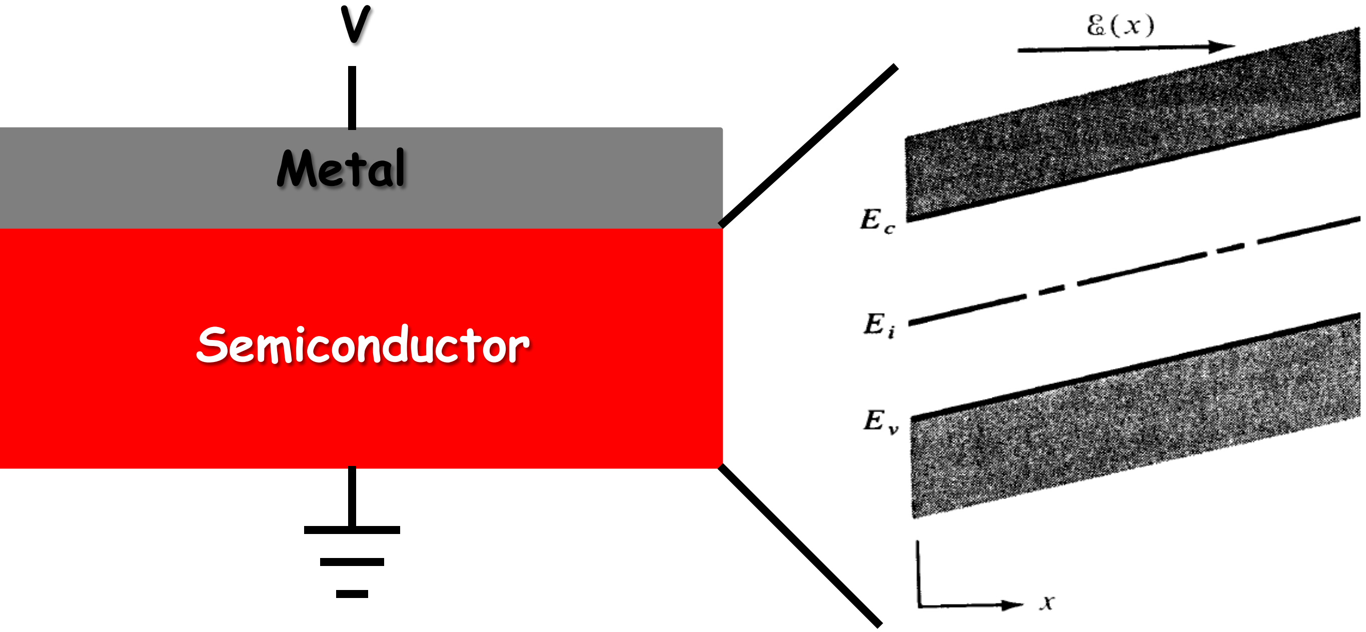


single MOS transistor

We will learn more about MOS and bipolar transistors later

Ideal Metal-Semiconductor Contacts

We talk a lot about semiconductors, but how do we **contact** them?



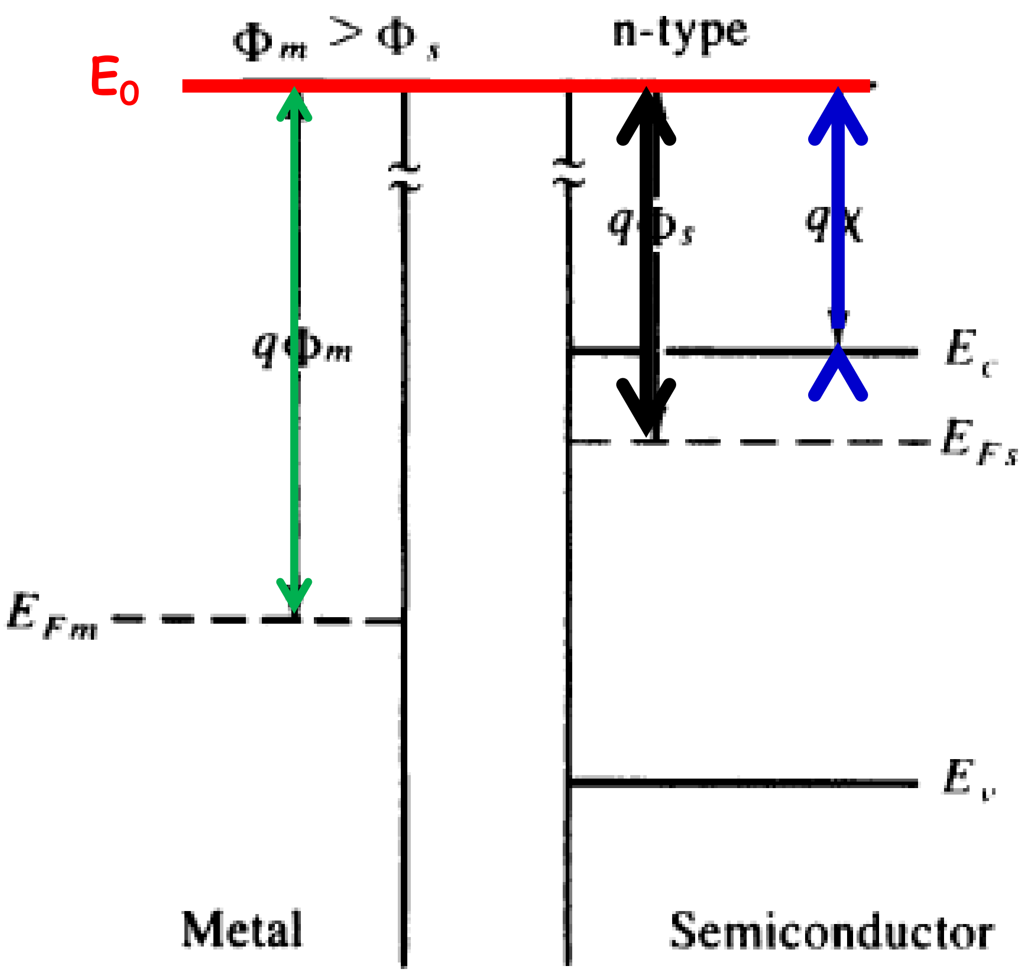
We have talked about the effects of electric fields, but how we apply one?

Use a **metal contact**…

In the **ideal case**, we assume:

- The metal and semiconductor are in intimate contact on the atomic scale with no layers of any type between the components.
- There is no interdiffusion or intermixing of the metal and the semiconductor.
- There are no adsorbed impurities or surface charges at the MS interface.

What do the band diagrams look like?

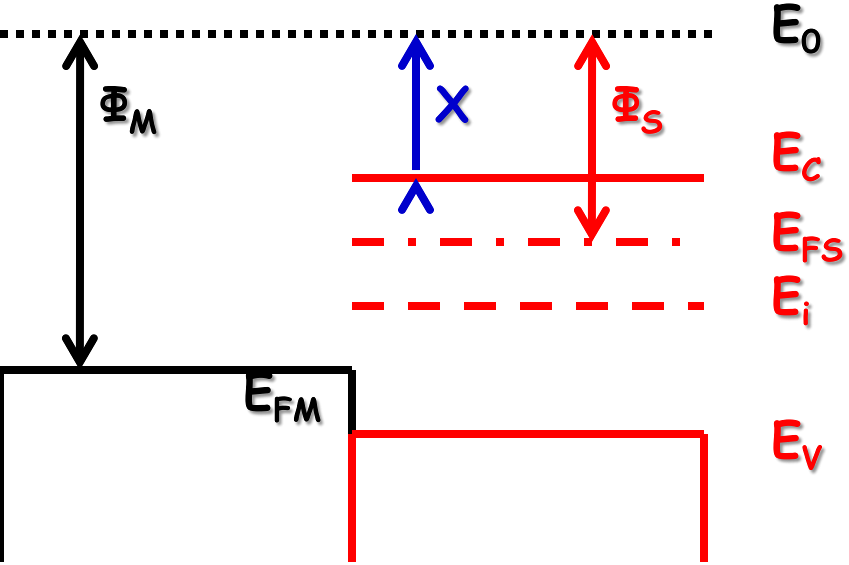


We need to understand several key energies in the metal and the semiconductor…

- The topmost energy is the **vacuum level**, E_0.
- The difference between the Fermi energy and the vacuum level is the **workfunction, Φ**.
  - This is a material property of the metal.
- The **semiconductor workfunction** is comprised of two properties.
  - The **electron affinity, X**.
  - $E_c - E_f$ which is a function of doping.

$$\Phi_S = \chi + \left(E_C - E_F\right)_{FB}$$

Now let's bring the metal and semiconductor together…

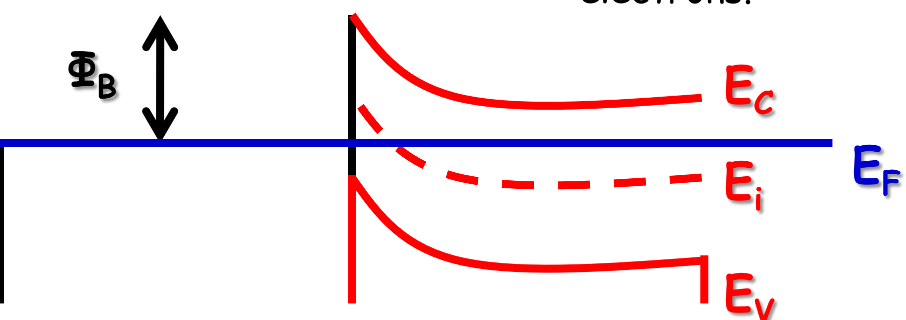


$\Phi_M > \Phi_S$

- When the materials are brought into contact with one another, they are not in equilibrium ($E_{FS} \neq E_{FM}$).

$$\Phi_B = \Phi_M - \chi \longrightarrow$$ Surface potential energy barrier for electrons.

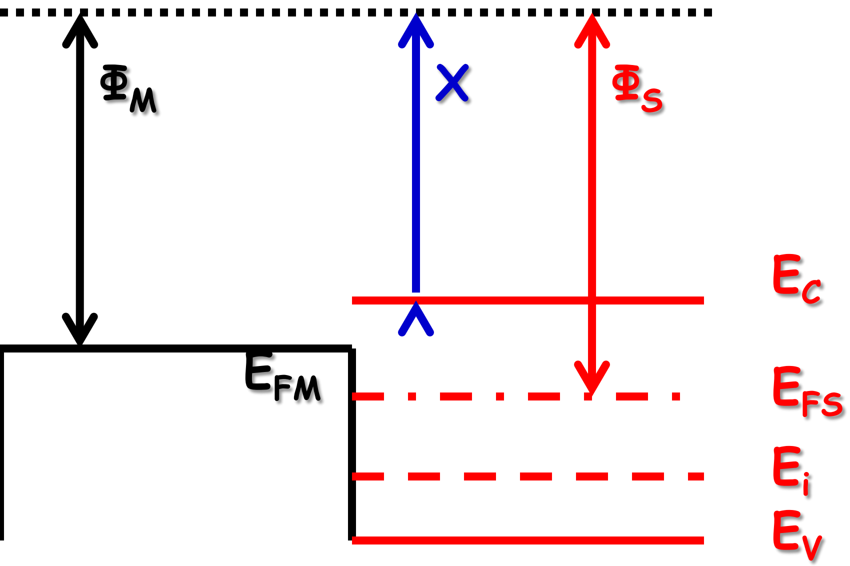
- Electrons begin moving from the semiconductor to the metal.



- The net transfer of electrons leaves a reduced electron concentration in the semiconductor and the barrier between the materials grows.
- Process continues until Fermi level is constant.

Ideal Metal-Semiconductor Contacts

What happens if $\Phi_M < \Phi_S$?



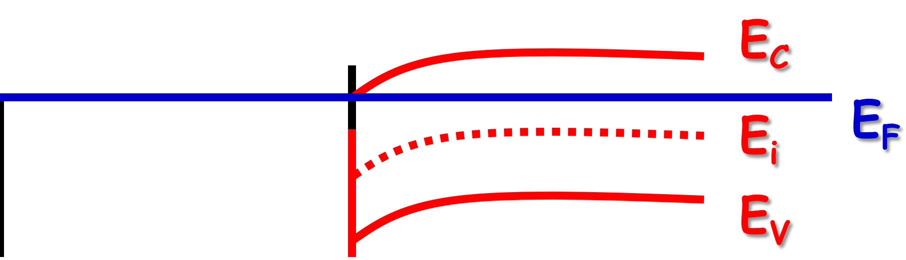
$\Phi_M < \Phi_S$

- When the materials are brought into contact with one another, they are not in equilibrium ($E_{FS} \neq E_{FM}$).

- Electrons begin moving from the metal to the semiconductor.

- The net transfer of electrons from the metal into the semiconductor leaves a net excess of electrons at the surface.

- Process continues until Fermi level is constant.

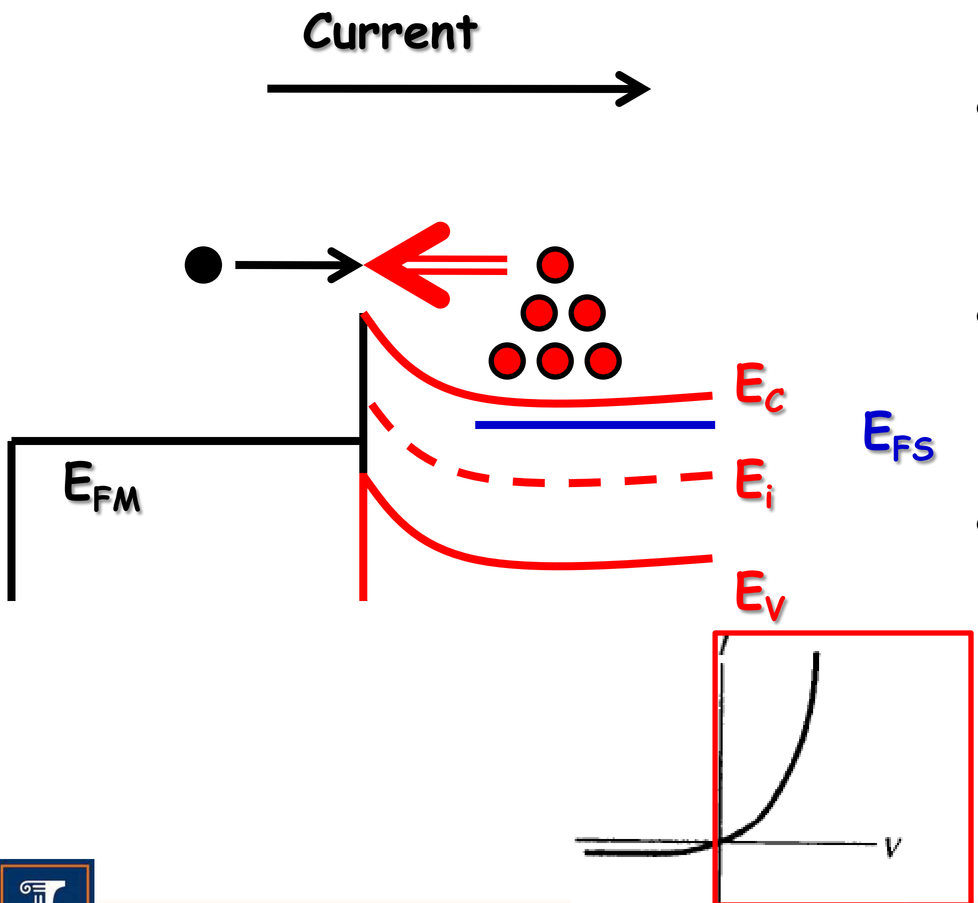


Ideal Metal-Semiconductor Contacts

But the point of adding contacts was to apply fields, let's look at this…

Apply positive bias, V…

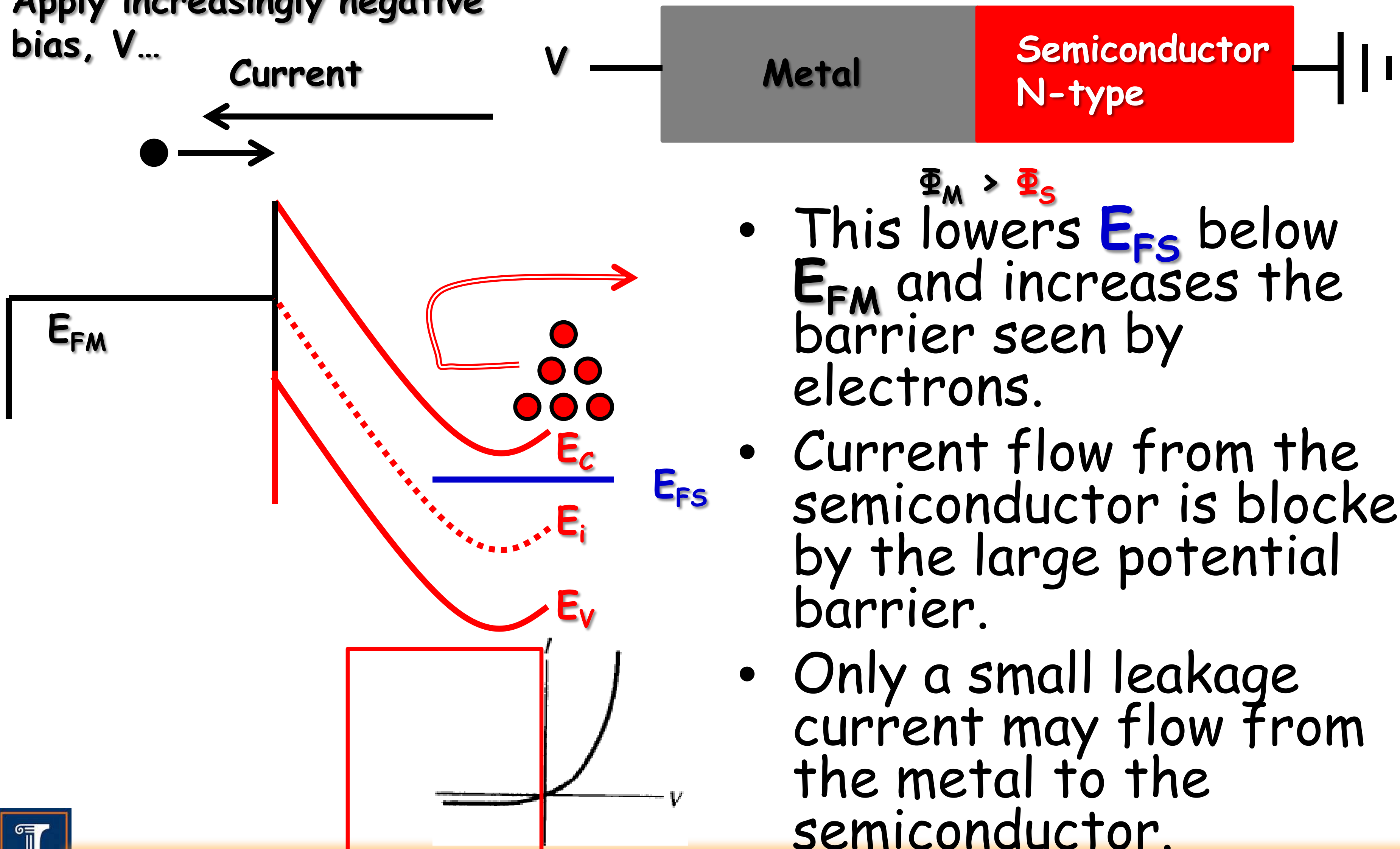
$\Phi_M > \Phi_S$



- This lowers E_{FM} below E_{FS} and reduces the barrier seen by electrons.
- Current begins to flow from the semiconductor to the metal.
- Continue to raise the positive bias and more electrons will have enough energy to surmount the barrier and contribute to current flow.

What happens if we apply a negative bias to the contact…

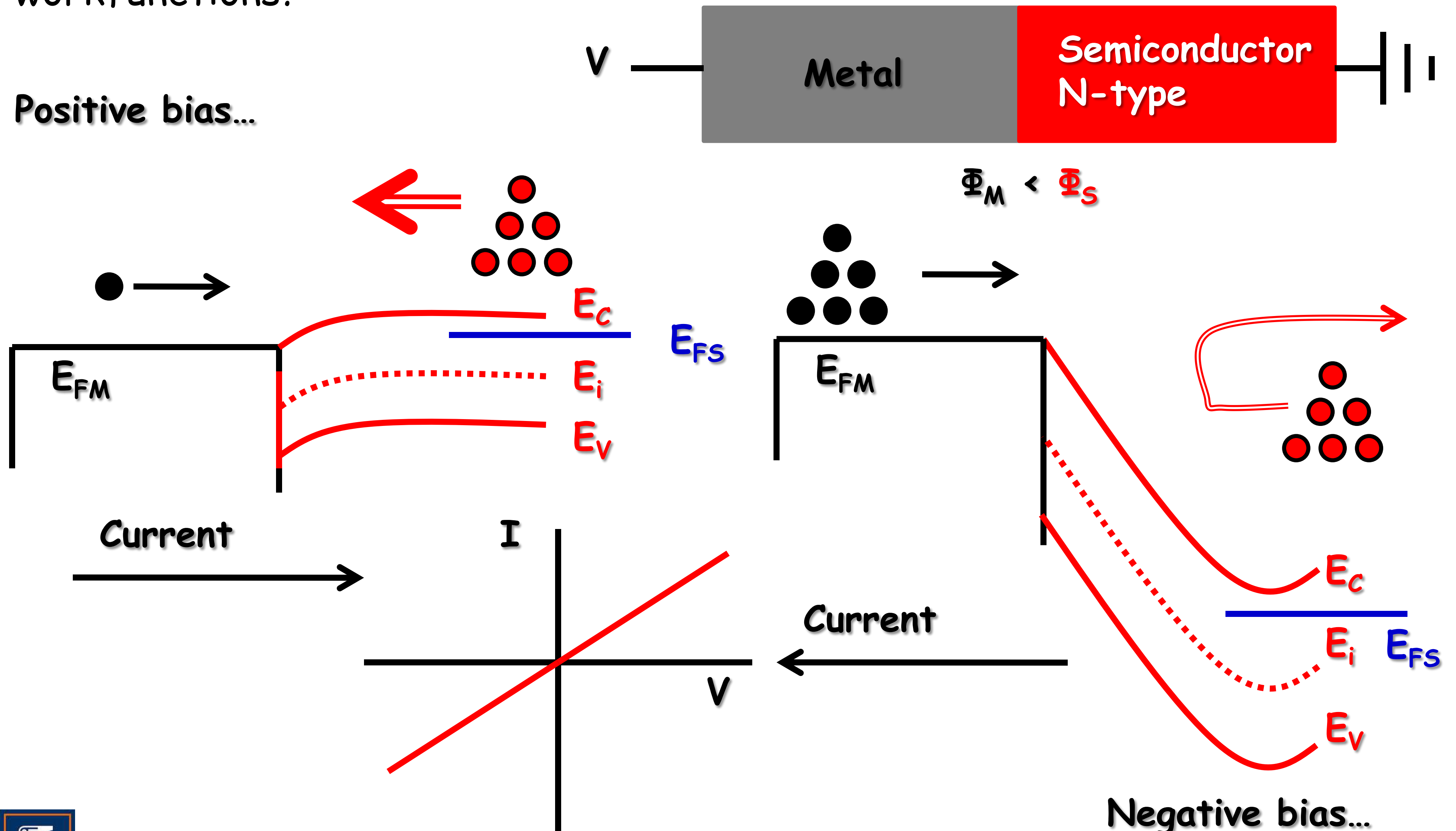
Apply increasingly negative bias, V…



- This lowers E_{FS} below E_{FM} and increases the barrier seen by electrons.
- Current flow from the semiconductor is blocked by the large potential barrier.
- Only a small leakage current may flow from the metal to the semiconductor.

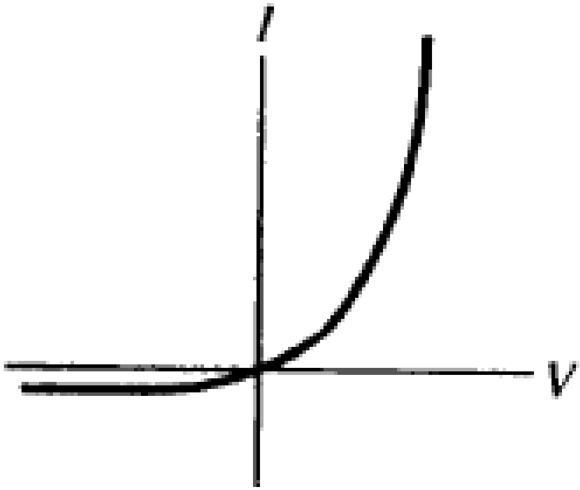
Ideal Metal-Semiconductor Contacts

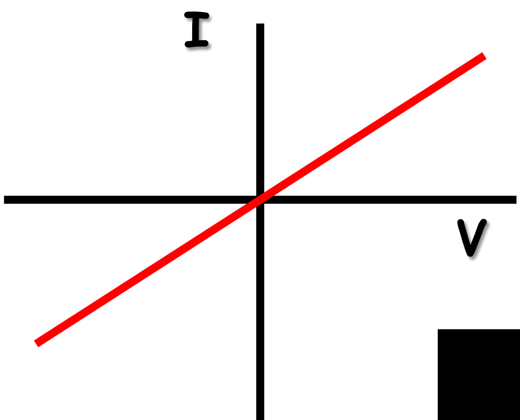
What happens when we reverse the relationship between the workfunctions?



Ideal Metal-Semiconductor Contacts

Let's summarize what we have so far…

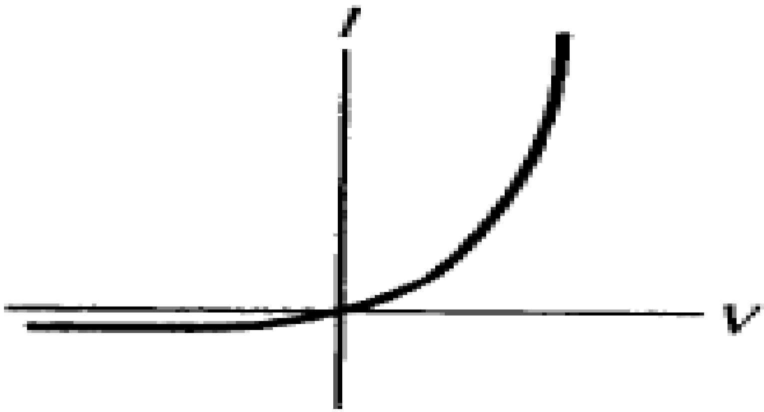


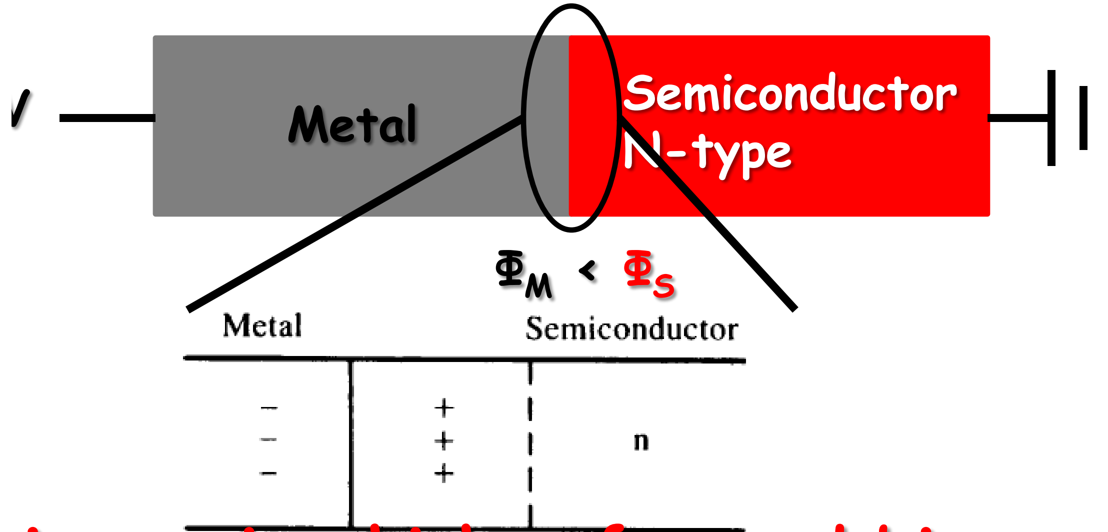


	N-type semiconductor	P-type semiconductor
$\phi_M > \phi_S$	Rectifying	Ohmic
$\phi_M < \phi_S$	Ohmic	Rectifying

Rectifying Contacts

When $\Phi_M < \Phi_S$ in an n-type seimconductor the contact is called **rectifying**…

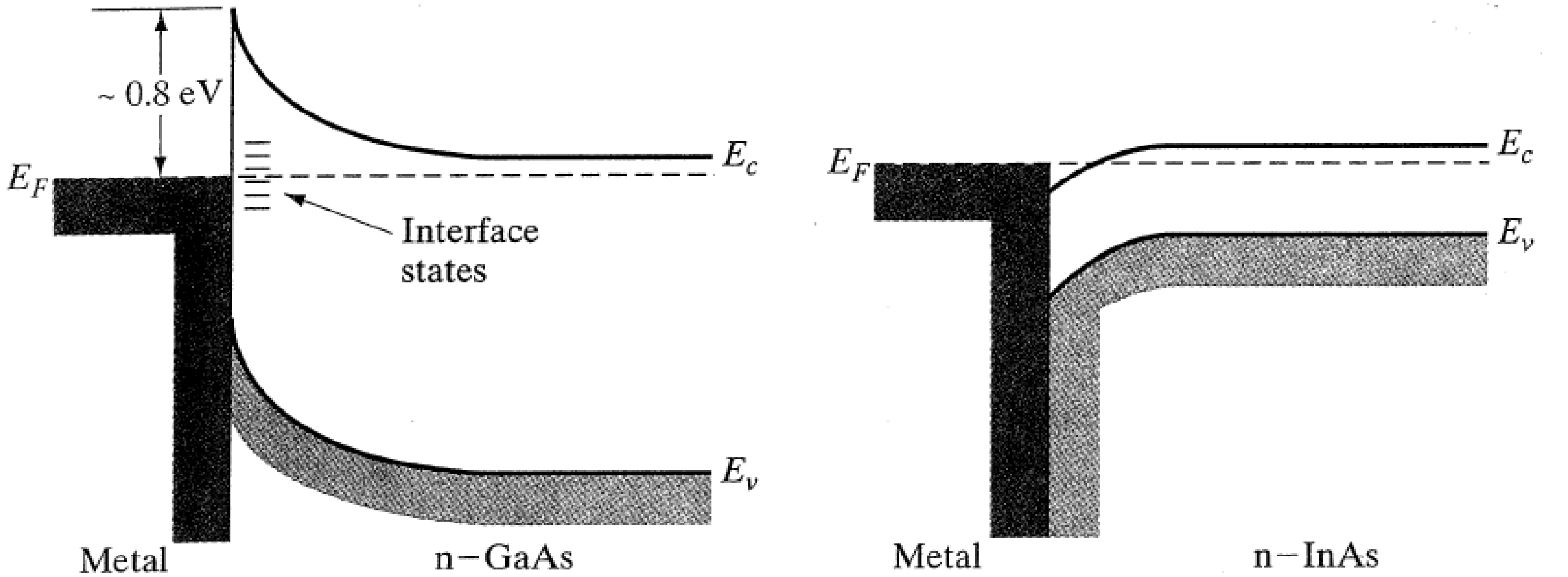




- **A rectifying contact is one in which a forward bias drives a large current but a reverse bias results in a small current.**
- Despite efforts, contacts are not ideal.
- In Si, exposure to air causes SiO_2 to form before the metal can be deposited. Something similar happens in GaAs too.
- Surface charges also change the surface potential barrier leading to unexpected behavior.

Rectifying Contacts

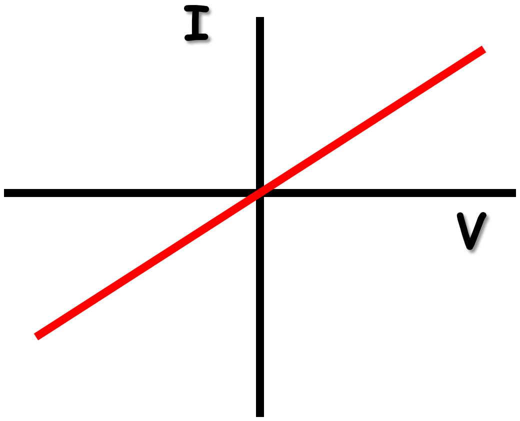
Surface states cause problems in III-V semiconductors…

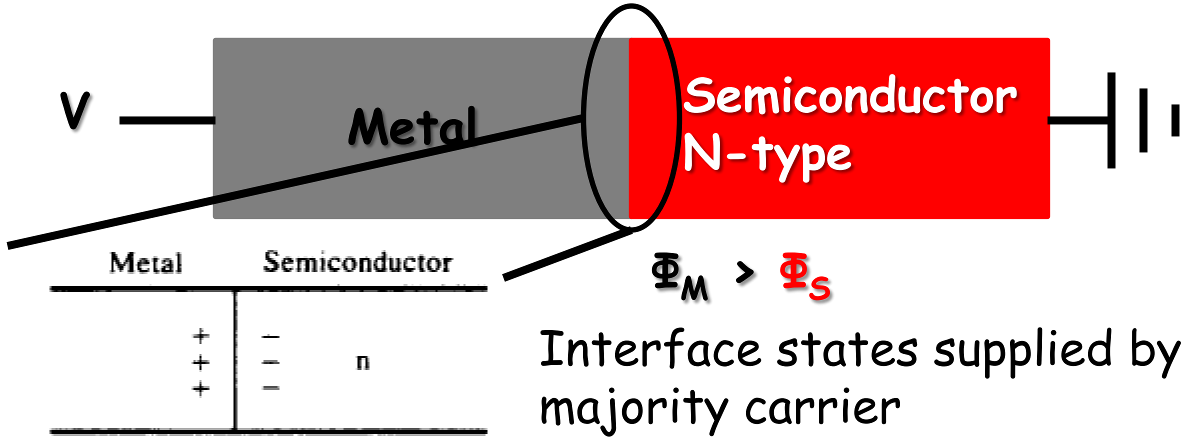


- Interface states pin the Fermi level at a fixed position regardless of the contact metal.
- Schottky barrier determined by surface states rather than metal and semiconductor workfunction difference.
- Effect is different in InAs as any metal becomes ohmic.

Ohmic Contacts

The other type of contact is **Ohmic**…



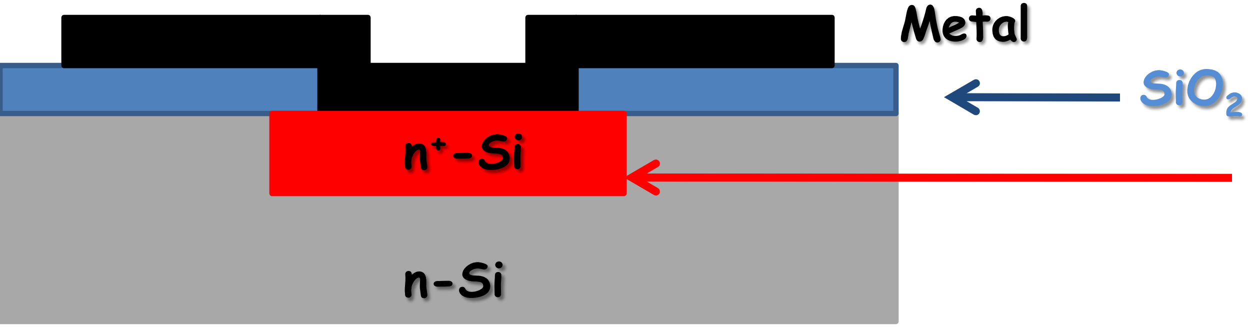


Interface states supplied by majority carrier

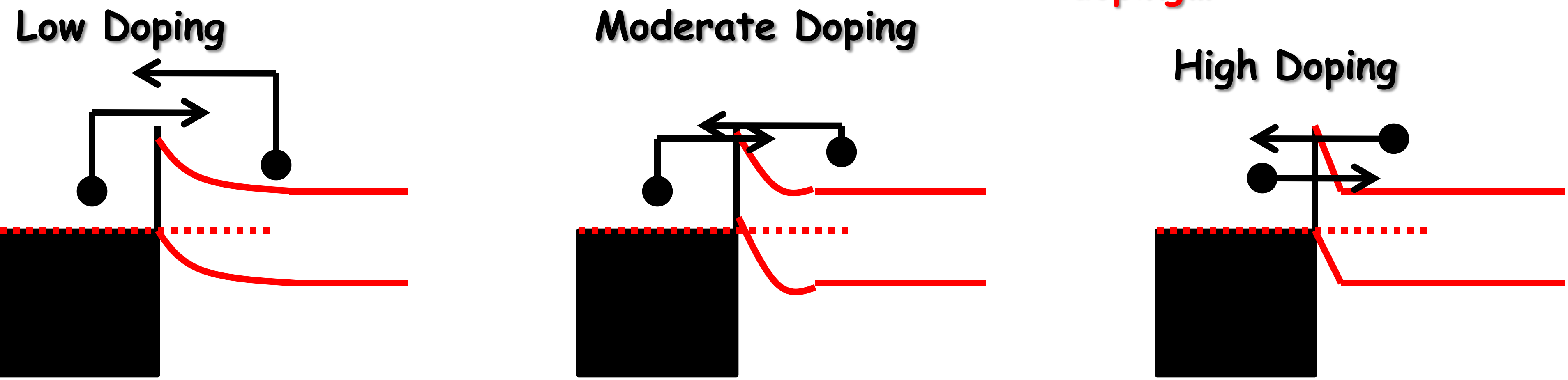
- **Ohmic contacts are low impedance contacts that allow current flow regardless of the polarity of the bias.**
- These are very important types of contact, so how do we make them?
- We know that surface states at the interface can cause significant problems and make all contacts rectifying.

Ohmic Contacts

How do you make an **ohmic** contact?



To make an ohmic contact to silicon, we need to use clever doping…



- Low doping – all thermionic emission.
- Moderate doping – some thermionic emission and some field emission.
- High doping – Mostly field emission.